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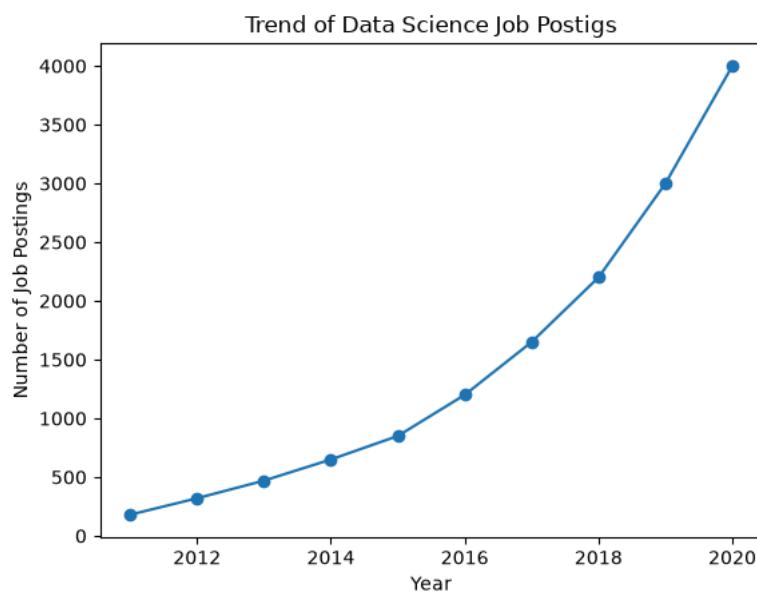
Register number : 250701573

Experimnet name : Exploratory data analysis to develop deep insights from a dataset

1)a.Analyze the trend of data

```
import pandas as pd
import matplotlib.pyplot as plt
data = {
    "Year": [2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020],
    "Job_Postings": [180, 320, 470, 650, 850, 1200, 1650, 2200, 3000, 4000]
}
df = pd.DataFrame(data)
print(df)
plt.plot(df["Year"], df["Job_Postings"], marker="o")
plt.title("Trend of Data Science Job Postigs")
plt.xlabel("Year")
plt.ylabel("Number of Job Postings")
plt.show()
```

	Year	Job_Postings
0	2011	180
1	2012	320
2	2013	470
3	2014	650
4	2015	850
5	2016	1200
6	2017	1650
7	2018	2200
8	2019	3000
9	2020	4000



1)b. Distribution of data Science roles

```
import pandas as pd
import matplotlib.pyplot as plt

roles = {
    "Job_Role": [
        "Data Scientist",
        "Data Analyst",
        "Data Engineer",
        "Machine Learning Engineer",
        "Data Scientist",
        "Data Analyst",
        "Data Engineer",
        "Data Scientist",
        "Data Analyst",
        "Machine Learning Engineer",
        "Data Engineer",
        "Data Scientist"
    ]
}
```

```

}

df = pd.DataFrame(roles)

role_distribution = df["Job_Role"].value_counts()

print(role_distribution)

role_distribution.plot(kind="bar")

plt.title("Distribution of Data Science Job Roles")
plt.xlabel("Job Role")
plt.ylabel("Number of Job Postings")

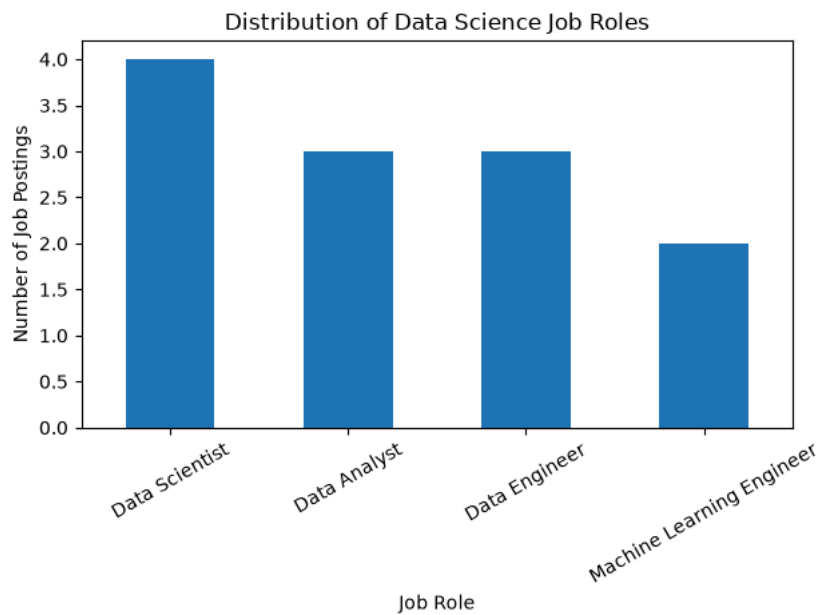
plt.xticks(rotation=30)
plt.tight_layout()
plt.show()

```

```

Job_Role
Data Scientist      4
Data Analyst        3
Data Engineer       3
Machine Learning Engineer  2
Name: count, dtype: int64

```



```
## 1)c. Structured.SemiStructured and UnStructured Data
```

```

import pandas as pd
import json

structured_data = {
    "ID": [101, 102, 103],
    "Name": ["Arun", "Priya", "Karthik"],
    "Department": ["CSE", "IT", "CSE"]
}

structured_df = pd.DataFrame(structured_data)

print("STRUCTURED DATA")
print(structured_df)

semi_structured_data = {
    "students": [
        {"ID": 101, "Name": "Arun", "Department": "CSE"},
        {"ID": 102, "Name": "Priya", "Department": "IT"},
        {"ID": 103, "Name": "Karthik", "Department": "CSE"}
    ]
}

print("\nSEMI-STRUCTURED DATA")
print(json.dumps(semi_structured_data, indent=4))

unstructured_data = """
Arun is a Computer Science student.
Priya is an Information Technology student.
Karthik is interested in Data Science.
"""

```

```
print("\nUNSTRUCTURED DATA")
print(unstructured_data)
```

STRUCTURED DATA

	ID	Name	Department
0	101	Arun	CSE
1	102	Priya	IT
2	103	Karthik	CSE

SEMI-STRUCTURED DATA

```
{
  "students": [
    {
      "ID": 101,
      "Name": "Arun",
      "Department": "CSE"
    },
    {
      "ID": 102,
      "Name": "Priya",
      "Department": "IT"
    },
    {
      "ID": 103,
      "Name": "Karthik",
      "Department": "CSE"
    }
  ]
}
```

UNSTRUCTURED DATA

Arun is a Computer Science student.
 Priya is an Information Technology student.
 Karthik is interested in Data Science.

1)d. Encryption and Decryption using Fernet

```
from cryptography.fernet import Fernet
```

```
key = Fernet.generate_key()
```

```
f = Fernet(key)
```

```
data = "My secret data"
```

```
encrypted = f.encrypt(data.encode())
```

```
decrypted = f.decrypt(encrypted).decode()
```

```
print("Original Data:", data)
```

```
print("Encrypted Data:", encrypted)
```

```
print("Decrypted Data:", decrypted)
```

```
Original Data: My secret data
```

```
Encrypted Data: b'gAAAAABqgcxTCMsPJb45penaT5wtQ4gkKpw8M7xo6xY-AyA1PPW5VitiOf0f4VLHxn-aBRyRLm02D3UuxZ9iQM2ODAMtYbnGrQ=='
```

```
Decrypted Data: My secret data
```

Start coding or [generate](#) with AI.